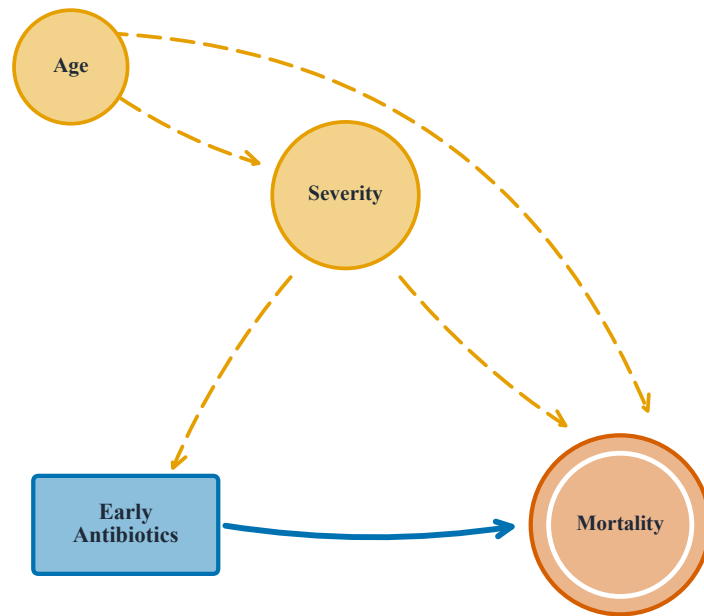


Causal DAG for the Three Critical Care Treatment Domains (identical structure, domain-specific coefficients)

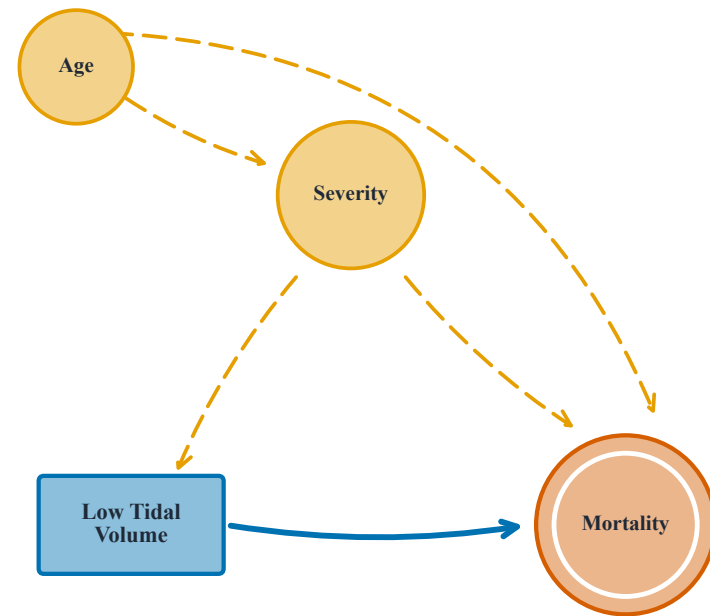
(A) Sepsis: Early Antibiotics



$T = \text{early antibiotic administration}$ | $Y = \text{in-hospital mortality (binary)}$

Minimal backdoor adjustment set $Z = \{\text{Severity}\}$

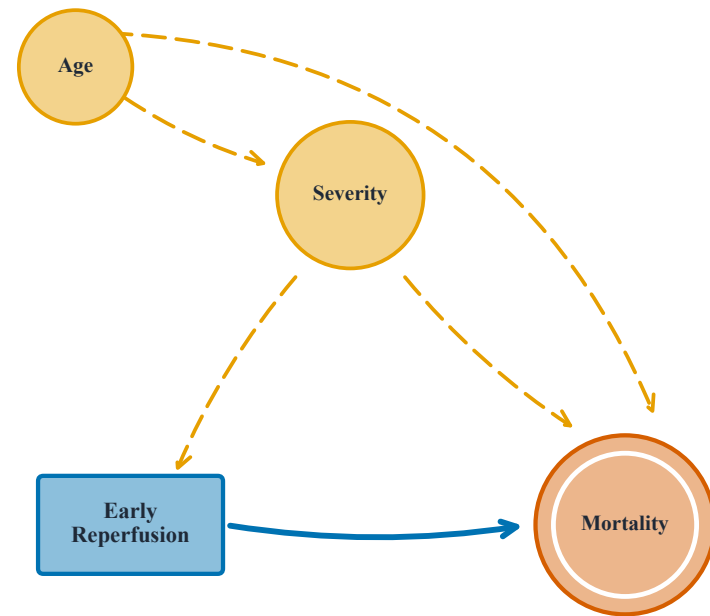
(B) ARDS: Low Tidal Volume



$T = \text{low tidal volume ventilation}$ | $Y = \text{in-hospital mortality (binary)}$

Minimal backdoor adjustment set $Z = \{\text{Severity}\}$

(C) ACS: Early Reperfusion



$T = \text{early reperfusion}$ | $Y = \text{in-hospital mortality (binary)}$

Minimal backdoor adjustment set $Z = \{\text{Severity}\}$

■ Treatment (T) — rectangle

■ Outcome (Y) — double circle

■ Baseline covariate — circle

— Treatment effect path

- - Backdoor (confounding) path